Activity 2. Equation-based design of photovoltaic module

Instructions: Read the next articles

1. Nguyen, X.H., Nguyen, M.P. Mathematical modeling of photovoltaic cell/module/arrays with tags in Matlab/Simulink. *Environ Syst Res* **4**, 24 (2015). https://doi.org/10.1186/s40068-015-0047-9.
2. Khatib, Tamer & Elmenreich, Wilfried. (2015). A Model for Hourly Solar Radiation Data Generation from Daily Solar Radiation Data Using a Generalized Regression Artificial Neural Network. International Journal of Photoenergy. 2015. 1-13. 10.1155/2015/968024
3. Saleem A, Liu N, Junjie H, Iqbal A, Waqar A. Comprehensive equation-based design of photovoltaic module to investigate its physical parameters and operating conditions used for small application. *Measurement and Control*. 2020;53(5-6):850-858.

In these articles, you will find the mathematical model for photovoltaic cells. Use a computational tool such as MATLAB to program this mathematical model and obtain the characteristic curves for Current versus Voltage (I-V) and Power versus Voltage (P-V) for different irradiance values (400 W/m², 600 W/m², 800 W/m², and 1000 W/m²); different temperature values (15°C, 25°C, 45°C, 70°C), and different shunt resistance values (5Ω, 10Ω, 100Ω, and 415Ω).

Send your codes and plots by email.